

MPLAB Harmony 2.0: From In-House To Open Source

The Microchip design community has released its latest software for microcontroller code development, MPLAB Harmony. Owing to ease of portability and compatibility to navigate on any Internet browser, MPLAB is on its way to becoming an essential firmware for designers.

Earlier, MPLAB was designed in-house by Microchip itself, but its recent release has been developed using open source NetBeans IDE by Oracle. Thus, the new version has a set of libraries, drivers and system services that can be easily accessed for developing applications. MPLAB provides integrated development environment for embedded microcontrollers and digital signal controllers.

The significance of MPLAB Harmony

MPLAB is a flexible software that integrates firmware development platform for PIC32 microcontrollers and provides a framework of software modules. These frameworks are easy to use and can be configured at designers' end as per their requirements. The flexibility to work in the real-time operating system (RTOS) adds to the value of MPLAB.

This open source firmware developer has peculiar features such as TCP/IP network stack and Wi-Fi support, fixed-point DSPk math libraries, graphics library, cryptographic library, peripheral

library, file system, system devices, drivers, operating system abstraction layer and MPLAB Harmony Configurator tool.

Features of MPLAB

MPLAB IDE software program has both in-built components and plug-in modules to configure

well as supporting modules such as IP, ICMP and DHCP.

The system is fully dynamic, supporting multiple interfaces such as Ethernet and Wi-Fi. Similar to other electronic hardware/software, benchmarking is carried out. The run-time in MPLAB benchmarking is done using the industry-standard tool Iperf.

The software library of MPLAB consists of MRF24WG0MA and MRF24WN0MA modules, which either create an 802.11 Wi-Fi network or join an existing 802.11 Wi-Fi network.

USB device stack.

MPLAB provides a user-friendly firmware to not only develop the software module framework but also a variety of USB devices. The IDE supports various

USB device classes with options of full speed and high operating speed.

Similarly, the USB host layer in MPLAB IDE supports multi-configuration and composite USB devices. Its elements can operate in multiple windows. For example, the modular architecture of USB device stack allows multiple USB operations in one single application.

Fixed-point DSPk math libraries.

Users need to develop algorithms for digital signal processing, which are supported by MPLAB library formats Q15 and Q31. Thus, the library makes functions such as complex math, vector math, matrix math, digital filters and transforms

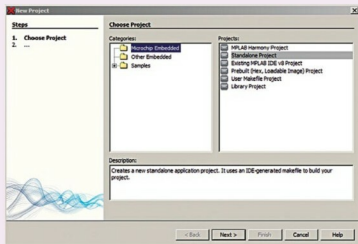


Fig. 1: New project window (Source: www.theengineertutor.com)

systems for various software and hardware tools. It allows users to create and edit source code, assemble, compile and link the source code, and view variables. Its features are described below.

TCP/IP network stack and Wi-Fi support.

The interaction between the physical network port and the user's application is supported by MPLAB Harmony TCP/IP stack. Thus, it includes modules for several commonly used application layers such as SMTP for sending e-mails, Telnet, serial-to-Ethernet and HTTP for webpage server. It also includes light-weight implementations such as TCP and UDP transport layers, as